Lecture 18: Section 5.4

Sigma Notation and Properties

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What is Sigma Notation?

Definition (Sigma Notation)

Sigma notation is a compact way to write sums:

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \cdots + a_n$$

- Σ is the capital Greek letter sigma (means "sum")
- i is the **index of summation**
- $i = 1$ is the **lower limit**
- n is the **upper limit**
- a_i is the **general term**

Write out the sum: $\sum_{k=1}^{5} k^2$

$$\sum_{k=1}^{5} k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$

Examples with Sigma Notation

Example

Write out the sum: $\sum_{i=0}^{3}(2i+1)$

Solution

$$\sum_{i=0}^{3}(2i+1) = (2\cdot 0+1)+(2\cdot 1+1)+(2\cdot 2+1)+(2\cdot 3+1) = 1+3+5+7 = 16$$

Example

Write using sigma notation: $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}$

Solution

$$\sum_{k=1}^{5}\frac{1}{k}$$

More Sigma Notation Examples

Example

Write using sigma notation: $2 + 4 + 6 + 8 + 10$

Solution

$$\sum_{i=1}^{5} 2i$$

Example

Evaluate: $\sum_{j=1}^{4}(j^2 - 1)$

Solution

$$\sum_{j=1}^{4}(j^2 - 1) = (1^2 - 1) + (2^2 - 1) + (3^2 - 1) + (4^2 - 1) = 0 + 3 + 8 + 15 = 26$$

More Sigma Notation Examples

Example

Write in expanded form: $\sum_{m=1}^{3} \frac{m}{m+1}$

Solution

$$\sum_{m=1}^{3} \frac{m}{m+1} = \frac{1}{2} + \frac{2}{3} + \frac{3}{4}$$

Example

Write using sigma notation: $\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9}$

Solution

$$\sum_{k=1}^{4} \frac{1}{2k+1}$$

Practice: Sigma Notation (10 problems)

1. Write out $\sum_{i=1}^{6} i$
2. Write out $\sum_{k=2}^{5} (k-1)^2$
3. Write $3+6+9+12+15$ in sigma notation
4. Write $\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\frac{4}{5}$ in sigma notation
5. Evaluate $\sum_{n=0}^{4} (n^3+1)$
6. Evaluate $\sum_{j=3}^{6} (2j-1)$
7. Write $1-2+3-4+5$ in sigma notation
8. Write $\sum_{i=1}^{4} (-1)^{i+1} i$ in expanded form
9. Evaluate $\sum_{k=1}^{5} \frac{k}{k+2}$
10. Write $1^3+2^3+3^3+4^3+5^3$ in sigma notation

Changing the Index of Summation

The Idea

We can change the starting point and index variable without changing the sum.

Example

Express $\sum_{k=2}^{5}(k-1)^2$ with index starting at 1.

Solution

Let $j = k - 1$. Then when $k = 2$, $j = 1$; when $k = 5$, $j = 4$:

$$\sum_{k=2}^{5}(k-1)^2 = \sum_{j=1}^{4} j^2$$

Both equal $1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$.

More changing the Index of Summation Examples

Example

Express $\sum_{i=0}^{3}(2i+1)$ with index starting at 1.

Solution

Let $j = i + 1$. Then when $i = 0$, $j = 1$; when $i = 3$, $j = 4$:

$$\sum_{i=0}^{3}(2i+1) = \sum_{j=1}^{4}(2(j-1)+1) = \sum_{j=1}^{4}(2j-1)$$

Both equal $1 + 3 + 5 + 7 = 16$.

Why Change Limits?

Reasons to Change Limits

1. To match a standard summation formula
2. To simplify calculations
3. To combine multiple sums
4. To prepare for limit calculations in area problems

Express $\sum_{k=3}^{7} k^2$ with index starting at 1.

Solution

Let $j = k - 2$. Then when $k = 3$, $j = 1$; when $k = 7$, $j = 5$:

$$\sum_{k=3}^{7} k^2 = \sum_{j=1}^{5} (j+2)^2$$

Check: $3^2 + 4^2 + 5^2 + 6^2 + 7^2 = 9 + 16 + 25 + 36 + 49 = 135$
$(1+2)^2 + (2+2)^2 + (3+2)^2 + (4+2)^2 + (5+2)^2 = 9 + 16 + 25 + 36 + 49 = 135$

Practice: Changing Limits (8 problems)

Change the index to start at 1:

1. $\sum_{k=0}^{4}(2k+3)$
2. $\sum_{i=2}^{6}(i-1)^2$
3. $\sum_{j=5}^{8} j^3$
4. $\sum_{n=-1}^{2}(3n+2)$
5. $\sum_{m=3}^{7} \frac{1}{m}$
6. $\sum_{k=10}^{13}(k-9)$
7. $\sum_{i=-2}^{1}(i^2+2i)$
8. $\sum_{j=4}^{9} \sqrt{j-3}$

Properties of Sigma Notation

Theorem (Properties of Sums)

Let c be a constant and let a_i and b_i be sequences. Then:

1. **Constant Factor:** $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$
2. **Sum:** $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$
3. **Difference:** $\sum_{i=1}^{n}(a_i - b_i) = \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i$

Example

Use properties to simplify $\sum_{i=1}^{5}(3i^2 + 2i)$.

Solution

$$\sum_{i=1}^{5}(3i^2 + 2i) = 3\sum_{i=1}^{5} i^2 + 2\sum_{i=1}^{5} i$$

Proof of Properties

Proof of Property 1: Constant Factor.

$$\sum_{i=1}^{n} ca_i = ca_1 + ca_2 + \cdots + ca_n = c(a_1 + a_2 + \cdots + a_n) = c\sum_{i=1}^{n} a_i$$

□

Proof of Property 2: Sum.

$$\sum_{i=1}^{n}(a_i + b_i) = (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n)$$

$$= (a_1 + a_2 + \cdots + a_n) + (b_1 + b_2 + \cdots + b_n) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$$

□

Examples Using Properties

Example

Simplify: $\sum_{k=1}^{n}(4k^3 - 2k + 5)$

Solution

$$\sum_{k=1}^{n}(4k^3 - 2k + 5) = 4\sum_{k=1}^{n} k^3 - 2\sum_{k=1}^{n} k + \sum_{k=1}^{n} 5$$

Example

Simplify: $\sum_{i=1}^{m}(2a_i + 3b_i - c_i)$

Solution

$$\sum_{i=1}^{m}(2a_i + 3b_i - c_i) = 2\sum_{i=1}^{m} a_i + 3\sum_{i=1}^{m} b_i - \sum_{i=1}^{m} c_i$$

More Examples

Example

Evaluate: $\sum_{i=1}^{10}(2i+3) - \sum_{i=1}^{10}(i-1)$

Solution

$$\sum_{i=1}^{10}(2i+3) - \sum_{i=1}^{10}(i-1) = \sum_{i=1}^{10}[(2i+3)-(i-1)] = \sum_{i=1}^{10}(i+4)$$

Special Sum: Constant Term

Theorem (Sum of a Constant)

$$\sum_{i=1}^{n} c = c + c + \cdots + c = n \cdot c$$

Example

Evaluate: $\sum_{i=1}^{100} 5$

Solution

$$\sum_{i=1}^{100} 5 = 100 \cdot 5 = 500$$

Example

Simplify: $\sum_{k=1}^{n}(3k^2+2k+7)$

Solution

$$\sum_{k=1}^{n}(3k^2+2k+7) = 3\sum_{k=1}^{n}k^2 + 2\sum_{k=1}^{n}k + 7\sum_{k=1}^{n}1 = 3\sum_{k=1}^{n}k^2 + 2\sum_{k=1}^{n}k + 7n$$

Practice: Properties of Sums (10 problems)

Use properties to simplify:

1. $\sum_{i=1}^{n}(4i^3 + 3i^2 - 2i)$
2. $\sum_{k=1}^{m}(5a_k - 3b_k)$
3. $\sum_{j=1}^{10}(2j + 1) + \sum_{j=1}^{10}(j - 3)$
4. $\sum_{i=1}^{n} 7$
5. $3\sum_{k=1}^{n} k^2 - 2\sum_{k=1}^{n} k$
6. $\sum_{i=1}^{5}(i^2 + 2i) - \sum_{i=1}^{5}(i^2 - 1)$
7. $\sum_{k=1}^{n}(c \cdot a_k + d \cdot b_k)$
8. $\sum_{j=1}^{m}(2j^3 - 3j^2 + 4j - 5)$
9. $\sum_{i=1}^{100}(3i - 2) - \sum_{i=1}^{100}(i + 1)$
10. $\sum_{k=1}^{n}(a_k + b_k)^2$ (Hint: First expand)

Important Summation Formulas

Theorem (Summation Formulas)

1. $\displaystyle\sum_{i=1}^{n} i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$
2. $\displaystyle\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$
3. $\displaystyle\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$

Memory Tips

- Sum of first n integers: $\frac{n(n+1)}{2}$ (easy to remember)
- Sum of squares: $\frac{n(n+1)(2n+1)}{6}$
- Sum of cubes: $\left[\frac{n(n+1)}{2}\right]^2 = (\text{sum of integers})^2$

Proof of Sum of Integers Formula

Proof: Let $S = 1 + 2 + 3 + \cdots + n$
Write it forward: $S = 1 + 2 + 3 + \cdots + n$
Write it backward: $S = n + (n-1) + (n-2) + \cdots + 1$
Add the two equations: $2S = (1+n) + (2+n-1) + (3+n-2) + \cdots + (n+1)$
Each pair sums to $n+1$, and there are n pairs.
So $2S = n(n+1)$, which gives $S = \frac{n(n+1)}{2}$.

Example: Find $1 + 2 + 3 + \cdots + 100$

$$\sum_{i=1}^{100} i = \frac{100 \times 101}{2} = \frac{10100}{2} = 5050$$

Examples Using Summation Formulas

Example

Evaluate $\sum_{i=1}^{20} i^2$.

Solution

$$\sum_{i=1}^{20} i^2 = \frac{20(20+1)(2\cdot 20+1)}{6} = \frac{20\cdot 21\cdot 41}{6} = \frac{17220}{6} = 2870$$

Evaluate $\sum_{k=1}^{10}(k^3 - k^2)$.

Solution

$$\sum_{k=1}^{10}(k^3 - k^2) = \sum_{k=1}^{10} k^3 - \sum_{k=1}^{10} k^2 = \left[\frac{10\cdot 11}{2}\right]^2 - \frac{10\cdot 11\cdot 21}{6}$$

$$= 55^2 - \frac{2310}{6} = 3025 - 385 = 2640$$

More Examples with Formulas

Example

Evaluate $\sum_{i=1}^{n}(3i^2 + 2i - 1)$.

Solution

$$\sum_{i=1}^{n}(3i^2 + 2i - 1) = 3\sum_{i=1}^{n} i^2 + 2\sum_{i=1}^{n} i - \sum_{i=1}^{n} 1$$

$$= 3 \cdot \frac{n(n+1)(2n+1)}{6} + 2 \cdot \frac{n(n+1)}{2} - n$$

$$= \frac{n(n+1)(2n+1)}{2} + n(n+1) - n$$

Example

Evaluate $\sum_{j=1}^{50}(2j-5)$.

Solution

$$\sum_{j=1}^{50}(2j-5) = 2\sum_{j=1}^{50} j - \sum_{j=1}^{50} 5 = 2 \cdot \frac{50 \cdot 51}{2} - 50 \cdot 5$$

$$= 2550 - 250 = 2300$$

Practice: Summation Formulas (10 problems)

Evaluate using formulas:

1. $\sum_{i=1}^{30} i$
2. $\sum_{k=1}^{15} k^2$
3. $\sum_{j=1}^{12} j^3$
4. $\sum_{i=1}^{n} (4i - 3)$
5. $\sum_{k=1}^{n} (k^2 + 3k + 1)$
6. $\sum_{i=1}^{40} (i^2 - i)$
7. $\sum_{j=1}^{25} (2j^2 - 3j + 4)$
8. $\sum_{k=1}^{n} (k^3 - 2k^2 + k)$
9. $\sum_{i=1}^{100} (3i^2 + 2i - 5)$
10. $\sum_{k=1}^{n} (ak^2 + bk + c)$ where a, b, c are constants